



2025

J23(B)

BOOKLET NO.

211079

Civil Engineering
Paper – II

Time Allowed : Three Hours

Maximum Marks : 250

Medium : English

Type of Paper : Conventional /Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Keep in mind the word limit indicated in the question if any.
8. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
9. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
10. Neat sketches may be drawn, wherever required.

Note : Candidates will be allowed to use Scientific (Non-programmable type) calculators.

P.T.O.

**SECTION – A**

Q1. Write the answers of following questions in about **150** words. **(5×10=50)**

Q1. (a) Explain the criteria for selecting materials for low cost housing, mass housing and high-rise buildings. **10**

Q1. (b) Describe various methods of contouring and explain how contour maps are used in preparing a topographical map. **10**

Q1. (c) Describe in detail the masonry construction principles for brick, stone and block masonry. **10**

Q1. (d) Outline the components and functions of a railway turnout and crossing and explain their significance in railway track design. **10**

Q1. (e) Write a short note on detailed estimate, approximate estimate and specification writing. **10**

Q2. (a) Describe timber as a construction material with reference to its properties, defects, seasoning and preservation treatments. Also discuss its use in low cost and mass housing. **20**

Q2. (b) Explain in detail the modern methods of surveying used in civil engineering and discuss their applications in railway engineering. **20**

Q2. (c) Explain the principles of functional planning of buildings for residential and specific purposes. Also discuss the essential building code provisions. **10**

Q3. (a) Discuss the principles and methods of economic analysis in construction management. Explain how it supports decision making. **20**

Q3. (b) Explain the principles of highway alignment and provide detailed notes on the geometric design elements and standards used in road design. Also discuss the structure of flexible and rigid pavements along with their design methodology. **20**

Q3. (c) Discuss the need for surface and subsurface drainage in roads. Explain various drainage arrangements and the role of culverts. **10**



- Q4. (a) Explain project profitability analysis and describe the BOOT model of financial planning. Discuss simple toll fixation criteria. 20
- Q4. (b) Explain the construction methods and material standards for 20
- i) WBM roads
 - ii) Bituminous roads
 - iii) Cement concrete roads and
 - iv) Stabilized soil roads
- Also describe pavement distresses, drainage arrangements and the role of traffic surveys in highway planning.
- Q4. (c) Describe the stages of construction activity and the importance of scheduling. 10
-

SECTION – B

- Q5. Write the answers of following questions in about 150 words. (5×10=50)
- Q5. (a) Explain consumptive use of water. Discuss factors affecting consumptive use and any one method of its estimation. 10
- Q5. (b) Discuss the activated sludge process. Explain formation of floc, F/M ratio, MLSS, sludge age and efficiency. 10
- Q5. (c) Explain different types of aquifers with neat sketches. Discuss how aquitards and aquicludes influence groundwater flow and artesian conditions. 10
- Q5. (d) Explain radioactive wastes and their disposal with reference to the following : 10
- i) Sources and the types of radioactive waste
 - ii) Safe disposal and management techniques.
- Q5. (e) Explain the principles of coagulation, flocculation and sedimentation with neat diagrams. Describe the various factors influencing each process. 10
-



- Q6. (a)** Discuss the infiltration process with factors affecting it. Derive Horton's infiltration equation and explain its significance in hydrologic modelling. **20**
- Q6. (b)** Explain in detail the parameters used for sewage characterization : BOD, COD, DO, TOC, nitrogen, solids (suspended, dissolved, volatile and fixed). Describe sampling analysis, significance and interpretation of results for treatment plant design. **20**
- Q6. (c)** Explain the concept of reservoir capacity. Describe the different types of storage with neat sketches. **10**

-
- Q7. (a)** Describe the design principles and working of the following canal regulatory and cross drainage structures : **20**
- i) Head regulators
 - ii) Canal falls
 - iii) Aqueductus
 - iv) Metering flumes
 - v) Canal outlets

- Q7. (b)** Explain in detail the sources, effects on environment and engineering control methods of air pollution. Discuss in detail the various pollution acts in India. **20**

- Q7. (c)** Explain various types of spillways with neat diagrams. Discuss the criteria for selecting a spillway type for a dam site. **10**

-
- Q8. (a)** Write a detail note on canal design and hydraulic principles with reference to the following : **20**
- i) Distribution system for canal irrigation
 - ii) Canal capacity
 - iii) Canal losses
 - iv) Most efficient canal section
 - v) Lined canals

- Q8. (b)** Describe the construction and working of trickling filters and oxidation ponds. Compare their performance, loading rates, microbiological aspects, design considerations and suitability for small towns and rural areas. **20**

- Q8. (c)** Explain the methods of collection, segregation and transportation of solid waste in rural and urban areas. **10**
-